

A UNION OF TWO NON-DDMO COPIES OF $K_{3,3}$ IS DIFFERENCE DISTANCE MAGIC ORIENTABLE

ABSTRACT. Aceska, Arcila-Maya, Carlson, Marr, Parnes, Ryan, Schuerger and Vasquez close their paper on difference distance magic labelings by asking: *is it possible for a union of non-DDMOGs to be DDMO?* We answer yes. Let $\vec{D}$ be the orientation of $K_{3,3}$ with parts $A = \{0, 1, 2\}$, $B = \{3, 4, 5\}$ and arcs $0 \rightarrow 3$, $4 \rightarrow 0$, $5 \rightarrow 0$, $3 \rightarrow 1$, $1 \rightarrow 4$, $1 \rightarrow 5$, $3 \rightarrow 2$, $2 \rightarrow 4$, $2 \rightarrow 5$. Then $\vec{D} \sqcup \vec{D}$ carries the difference distance magic labeling $(6, 1, 5, 10, 2, 8)$, $(12, 3, 9, 11, 4, 7)$ of $\{1, \dots, 12\}$, while $K_{3,3}$ itself has no orientation admitting such a labeling. So a disjoint union of two graphs, *neither* of which is difference distance magic orientable, is difference distance magic orientable.

1. THE QUESTION

An *oriented graph* is a digraph with no pair of opposite arcs. Following Altman, Calzado, Freyberg, Kovář, Lewis, Marr, Ross and Via [2] and Aceska et al. [1], for a vertex v of an oriented graph $\vec{G}$ on n vertices and a labeling f of its vertices put

$$\text{wt}(v) = \sum_{u \rightarrow v} f(u) - \sum_{v \rightarrow u} f(u),$$

the in-label sum minus the out-label sum. A *difference distance magic* (DDM) labeling of $\vec{G}$ is a bijection $f: V(\vec{G}) \rightarrow \{1, \dots, n\}$ with $\text{wt}(v) = 0$ for *every* vertex v ; an oriented graph possessing one is a *DDMOG*, and an undirected graph is *DDMO* when at least one of its orientations is a DDMOG.

The source paper [1] ends with a paragraph of open questions. Its third question is unnumbered prose in the Conclusion (Section 13), in no theorem-like environment, and reads, verbatim:

For example, is it possible for a union of non-DDMOGs to be DDMO?

Its immediate context, verbatim, is: “*Furthermore, although most of the DDMOGs in this paper have been connected, disconnected DDMOGs are of interest as well. For example, is it possible for a union of non-DDMOGs to be DDMO?*” Throughout, “union” means disjoint union, which is the source’s own usage: its Theorem 5.1 writes a disconnected oriented graph as $\bigcup_i \vec{G}_i$.

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The question unions *oriented* objects and then applies the *undirected* predicate, so it has two inequivalent readings, and we answer both. Fix $k \geq 2$.

- (A) (orientations fixed throughout) There are oriented graphs $\vec{G}_1, \dots, \vec{G}_k$, none a DDMOG, with $\vec{G}_1 \sqcup \dots \sqcup \vec{G}_k$ a DDMOG.
 (B) (undirected) There are graphs $H_1, \dots, H_k$, none DDMO, with $H_1 \sqcup \dots \sqcup H_k$ DDMO.

An instance of (B) yields one of (A), because a disjoint union has no arcs between parts, so any DDMOG orientation of $H_1 \sqcup \dots \sqcup H_k$ restricts to orientations of the H_i , each a non-DDMOG. We require $k \geq 2$.

Theorem 1. *Reading (B), and hence reading (A), holds with $k = 2$: two disjoint copies of $K_{3,3}$ form a DDMO graph on 12 vertices, although $K_{3,3}$ is not DDMO.*

No minimality of the order 12 is claimed. The mechanism is one line: a component of a *disconnected* DDMOG on n vertices receives an arbitrary n_i -subset of $\{1, \dots, n\}$, whereas being a DDMOG in its own right demands the *interval* $\{1, \dots, n_i\}$.

Since wt is defined by an equality to zero, replacing wt by $-\text{wt}$ changes nothing below; this matters only because [1] reverses the usual superscript convention, so that its N^+ is the in-neighbourhood. All statements are in the convention displayed above.

2. THE WITNESS ON 12 VERTICES

Let $\vec{D}$ be the orientation of $K_{3,3}$ on $A = \{0, 1, 2\}$, $B = \{3, 4, 5\}$ with the nine arcs

$$\begin{aligned} 0 \rightarrow 3, \quad 4 \rightarrow 0, \quad 5 \rightarrow 0, \\ 3 \rightarrow 1, \quad 1 \rightarrow 4, \quad 1 \rightarrow 5, \\ 3 \rightarrow 2, \quad 2 \rightarrow 4, \quad 2 \rightarrow 5. \end{aligned}$$

Each of the nine pairs in $A \times B$ occurs exactly once and no pair occurs twice, so $\vec{D}$ is indeed an orientation of $K_{3,3}$; its in-degrees are $(2, 1, 1, 1, 2, 2)$ and its out-degrees $(1, 2, 2, 2, 1, 1)$.

For this orientation the six weight equations collapse. Reading the arcs off,

$$\begin{aligned} \text{wt}(0) &= f_4 + f_5 - f_3, & \text{wt}(1) &= \text{wt}(2) = f_3 - f_4 - f_5, \\ \text{wt}(3) &= f_0 - f_1 - f_2, & \text{wt}(4) &= \text{wt}(5) = f_1 + f_2 - f_0, \end{aligned}$$

so every row of the weight matrix is $\pm(f_0 - f_1 - f_2)$ or $\pm(f_3 - f_4 - f_5)$ and

$$(1) \quad \text{wt} \equiv 0 \iff f_0 = f_1 + f_2 \quad \text{and} \quad f_3 = f_4 + f_5.$$

Equivalently: the label sets realised by $\vec{D}$ are exactly the 6-sets of the form $\{p, q, p+q\} \cup \{r, s, r+s\}$.

Proposition 2. *Let $\vec{G}_{12} = \vec{D} \sqcup \vec{D}$, and label the first copy by $(f_0, \dots, f_5) = (6, 1, 5, 10, 2, 8)$ and the second by $(12, 3, 9, 11, 4, 7)$. This is a DDM labeling of $\vec{G}_{12}$.*

Proof. The two label sets $\{1, 2, 5, 6, 8, 10\}$ and $\{3, 4, 7, 9, 11, 12\}$ are disjoint with union $\{1, \dots, 12\}$, so the labeling is a bijection onto $\{1, \dots, 12\}$. By (1) all twelve weights vanish as soon as

$$1 + 5 = 6, \quad 2 + 8 = 10, \quad 3 + 9 = 12, \quad 4 + 7 = 11,$$

which is four additions. Equivalently, the witness is the partition $\{1, \dots, 12\} = (\{1, 5, 6\} \cup \{2, 8, 10\}) \sqcup (\{3, 9, 12\} \cup \{4, 7, 11\})$ into four triples $\{a, b, a+b\}$. $\square$

3. NEITHER COPY IS DDM-ORIENTABLE

Lemma 3 (Altman et al. [2], Theorem 5, at $m = n = 3$). *No orientation of $K_{3,3}$ has a DDM labeling. Hence $K_{3,3}$ is not DDMO and every orientation of $K_{3,3}$ is a non-DDMOG.*

This is not new: [2, Theorem 5] states that $K_{m,n}$ with $m, n \geq 3$ has an orientation permitting a DDM labeling if and only if $(m+n)(m+n+1) = 4s$ for some $s \in \mathbb{N}$, and $(3+3)(3+3+1) = 42$ is not a multiple of 4. That paper is the first item in the bibliography of [1]. We reproduce the argument, which is the necessity half of theirs, only so that the witness is self-contained.

Proof. Let $\vec{K}$ be any orientation of $K_{3,3}$ and let f be any labeling with all weights zero. Summing the weights over A : every arc of $\vec{K}$ joins A to B , so no $f(a)$ with $a \in A$ occurs in $\sum_{a \in A} \text{wt}(a)$, while $f(b)$ occurs with coefficient $\text{outdeg}(b) - \text{indeg}(b) = 2 \text{outdeg}(b) - 3$, which is odd because $\text{deg}(b) = 3$. Hence $0 = \sum_{a \in A} \text{wt}(a) \equiv \sum_{b \in B} f(b) \pmod{2}$, so $S_B := \sum_{b \in B} f(b)$ is even, and symmetrically S_A is even. But a DDM labeling has $S_A + S_B = 1 + 2 + \dots + 6 = 21$, which is odd; a contradiction. $\square$

Lemma 3 with Proposition 2 proves Theorem 1 and answers the question in reading (B). The union defeats the parity obstruction because each copy receives a 6-subset of $\{1, \dots, 12\}$ of even sum (32 and 46), whereas the interval $\{1, \dots, 6\}$ has odd sum 21.

4. RELATION TO THEOREM 5.1 OF THE SOURCE

The witness lies outside the hypothesis of Theorem 5.1 of [1], which asserts that $\vec{G} = \bigcup_{i=1}^k \vec{G}_i$ is a DDMOG if $\vec{G}_1$ is a DDMOG and $\vec{G}_2, \dots, \vec{G}_k$ are DDMOGs with $\text{imb}(\vec{G}_i) = 0$: here no component is a DDMOG at all, by Lemma 3. That theorem is stated as a sufficient condition only, and the source itself declines the case the witness inhabits — the prose introducing it reads “*However, if a union of DDMOGs happens to contain more than one component that is not balanced, it is not immediately clear if a DDM labeling will exist for this union.*”

5. VERIFICATION, AND WHAT IS NOT SETTLED

By hand. Everything above is hand-sized and needs no machine: the 12-vertex witness is four additions, and Lemma 3 is a parity count.

By machine. The accompanying program `verify.py` (Python 3.9+, standard library only), in exact integer and rational arithmetic, reads the objects printed above and re-checks them, and confirms Lemma 3 by two independent routes over all 512 orientations of $K_{3,3}$, which agree. Its recorded transcript also carries checks on further material — minimum orders for witnesses, order-5 enumerations, and a parity condition for k copies of $\vec{D}$ — none of which is claimed in this note and none of which anything above depends on; where that transcript names a lemma or a remark of the paper for a step it did not re-run, no such numbered statement appears above.

What is not settled.

- No theorem describing which orders admit witnesses is proved here, and no witnesses are counted: Theorem 1 exhibits one witness, on 12 vertices, and nothing here rules out a witness of smaller order.
- The published census of 6-vertex DDMOGs in [2] — “*there are 22 unique 6-vertex oriented graphs on two underlying non-isomorphic graphs with DDM labelings*” — is *not* reproduced here and no agreement with it is claimed. The only order-6 object this note uses is $K_{3,3}$, whose 512 orientations are swept exhaustively. Likewise the 7- and 8-vertex counts of [2] play no role.
- The ingredient the argument leans on is published, not new: Lemma 3 is [2, Theorem 5] at $m = n = 3$, and the proof reproduced here is the necessity half of theirs. What is claimed here is the answer to the question and the 12-vertex witness.
- Besides [1] and [2], the only further paper we located on difference distance magic labelings is [3], the one paper we found citing the source; its constructions all adjoin a neutralising edge set and so produce connected DDMOGs, and none of its own open questions restates or answers the one answered here. The words “disjoint”, “union”, “component” and “disconnect” do not occur in [2].
- The other two open questions of the same paragraph of [1] — which graphs are DDMO, and what properties distinguish DDMOGs from non-DDMOGs — remain open.
- Splitting an interval of labels into blocks of prescribed sum is the standard device of the undirected and group distance-magic literature (see e.g. [4]); the mechanism of the 12-vertex witness is recognisably that device, transplanted to the oriented setting.

REFERENCES

- [1] R. Aceska, C. Arcila-Maya, R. Carlson, A. Marr, P. Parnes, J. Ryan, H. Schuerger, and A. Vasquez, *New results on difference distance magic labelings*, arXiv:2601.20492v1,

2026. arXiv:2601.20492; doi:10.48550/arXiv.2601.20492. The question answered here is unnumbered prose in the Conclusion (Section 13) of that version, and is quoted in full in Section 1.

- [2] K. Altman, L. Calzado, B. Freyberg, P. Kovář, E. Lewis, A. Marr, L. Ross, and R. Via, *Difference distance magic oriented graphs*, Res. Math. Sci. **11** (2024), no. 4, art. 67. doi:10.1007/s40687-024-00482-7.
- [3] R. Aceska, T. Lort, and A. Ripperger, *Ornaments and difference distance magic oriented graphs*, arXiv:2606.16001v3, 2026. arXiv:2606.16001. Every DDMOG constructed there is connected, so none of its results bears on the question answered here.
- [4] B. Freyberg, *On constant sum partitions and applications to distance magic-type graphs*, AKCE Int. J. Graphs Comb. **17** (2020), no. 1, 373–379. doi:10.1016/j.akcej.2019.02.003.